

EM Billing System — Implementation Plan

Cart-based EM requests with PI approval · auto-enable or RSSD Jira · monthly per-PTA finance invoicing · carry-forward debt tracking and backcharge ledger. Merges the finalized plan with the workflow diagrams.

v1.0 · July 10, 2026 · Stanford REDCap · stanford/external_module_manager_v9.9.9

Companions: *RMA_Migration_Plan_and_Open_Questions.html* (decision history) ·

Workflow_Diagrams.html v2.0 (standalone diagrams) · *EM_Billing_Mockup_Prompt.md* (UI mockups spec)

1) Key Design Decisions Finalized July 10, 2026

Request flow

- **Catalog & cart** page in each project's EM menu (som-prod v1, Design right required); catalog, fees, and a new per-EM `auto_enable` flag live as records in the EM-records project (PID 16000).
- Checkout validates the **PTA** (regex) and pre-creates a record in a dedicated **request-tracking REDCap project**, then redirects to its survey (no data in the URL).
- Requester picks the **PI/Admin approver** on the survey via the Stanford person-lookup EM; a REDCap Alert emails the PI the approval survey.
- PI approves/rejects the **whole request** (PTA + all EMs, one decision).
- On approval (`redcap_save_record` hook): PTA syncs into `pta_accounts` + `project_pta_mapping` ; flagged EMs auto-enable (failure → `approval_error` , retryable); the rest get **one RSSD Jira ticket per request** (REST API), `awaiting_super_user` until reconciliation sees the manual enable.
- The planned `em_enable_requests` / `em_enable_request_items` entities are **not built** — tracking-project records supersede them.

Billing, invoice & debt

- **Line-level storage; grouping is a view.** `finance_billing_lines` (one per project+EM+month, PTA snapshotted) under `finance_billing_runs` ; run view groups PTA → project → EM. Replaces the JSON-snapshot `monthly_pta_reports` .

- **Debt owner = PTA.** Outstanding Balances ledger keyed by PTA — open rejected/hold lines attributed to each project's *current* PTA (snapshot kept for audit).
- **Finance rejects per PTA.** Editable CSV = one row per PTA; import fans status out to that PTA's pending lines. Read-only per-line detail CSV alongside.
- **Carry-forward** re-bills the project's *current* PTA — same amount, attempt+1, `carried_from_line_id` ; created regardless of current enablement (already-incurred debt).
- **Unmapped projects** → lines at `hold` + "No PTA mapping" (never dropped; carry forward until mapped).
- **Manual overrides** any time (incl. post-finalize) with required audited note — closes settled backcharges.
- **Scope v1: som-prod only** (`instance = som-prod`); multi-instance keys deferred.

Deactivation & reconciliation

- Module-injected UI on REDCap's **core EM manager page**: per-EM fee/PTA/status annotations + a module-owned **Disable** button (required reason, immediate, audited). Re-enable = full new request.
- **Nightly reconciliation cron**: out-of-band disables → `auto_detected` acknowledgment queue; Jira-resolved manual enables → line flipped to `enabled` .
- **Go-live baseline**: one-time seeding of *grandfathered* tracking records for every currently-enabled billable EM.

Security fix (must ship first)

`ajax/finance_report_action.php` and `ajax/finance_report_download.php` are currently in `no-auth-pages` with **no permission or token check** — remove them from `no-auth-pages` and guard super-user-only (page + both endpoints).

2) Delivery Milestones

Mockups — leadership review `EM_Billing_Mockup_Prompt.md`

Clickable HTML mockups under `mockups/` : catalog/cart, PI approval survey, billing runs, CSV import, outstanding balances, request admin + core-page strip. No backend; consistent status-badge system.

Milestone 0 — Design docs

Update *RMA_Migration_Plan_and_Open_Questions.html* (Section 9 revision) and *Workflow_Diagrams.html* (v2.0) — `done July 10, 2026` .

Milestone 1 — Billing engine + finance CSV loop

New `finance_billing_runs` / `finance_billing_lines` entities; drop `monthly_pta_reports`.
`generateBillingRun()` (duplicate gate · new lines from `external_modules_charges` · carry-forward · unmapped→hold), summary/detail CSV exports, all-or-nothing import with per-PTA fan-out, manual overrides, PTA ledger. Rework `ajax/finance_report_action.php`, `ajax/finance_report_download.php`, `pages/finance_reports.php` tabs (Billing Runs · Generate · Import CSV · Outstanding Balances · PTA Accounts · Mapping). Includes the no-auth security fix.

Milestone 2 — Request flow

Request-tracking project (instruments: request survey · approval survey · repeating EM line) + Alert; PID 16000 `auto_enable` / `requestable` fields; `pages/em_catalog.php` + `ajax/em_request_action.php`; `redcap_save_record` hook (PTA sync · auto-enable · Jira REST); `pages/request_admin.php` (requests · retries · grandfather seeding). New system settings: `em-request-project-id`, `pta-regex`, `jira-*`.

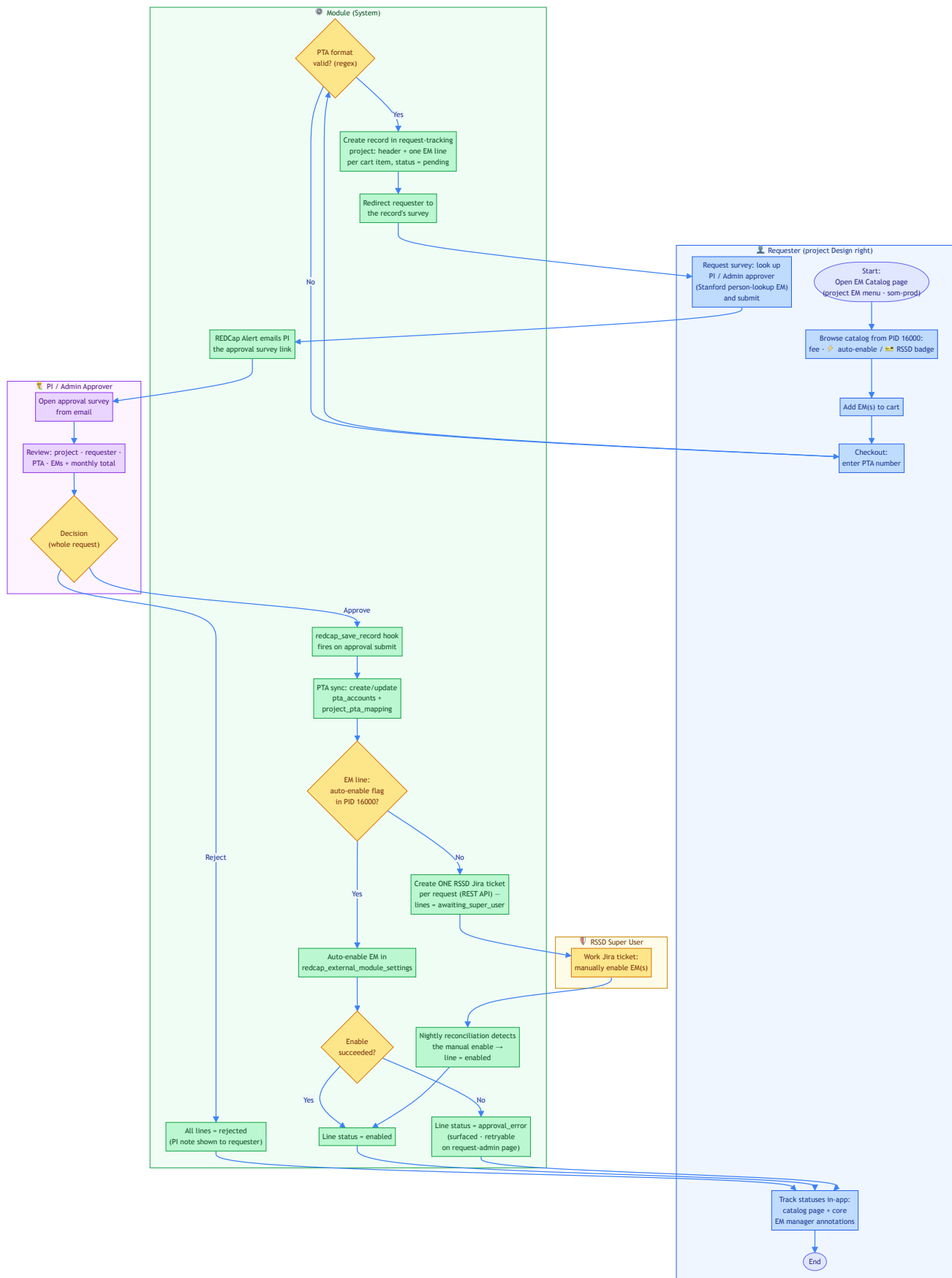
Milestone 3 — Deactivation, core-page injection, reconciliation

`redcap_every_page_top` injection on the core EM manager page (annotations + Disable); deactivate action; nightly `em_reconciliation` cron (auto-detect queue · Jira-line closure).

Out of scope here: Phase D cutover sequencing and Phase E R2P2 historical migration — planned separately (Migration Plan §8 holds the migration decisions).

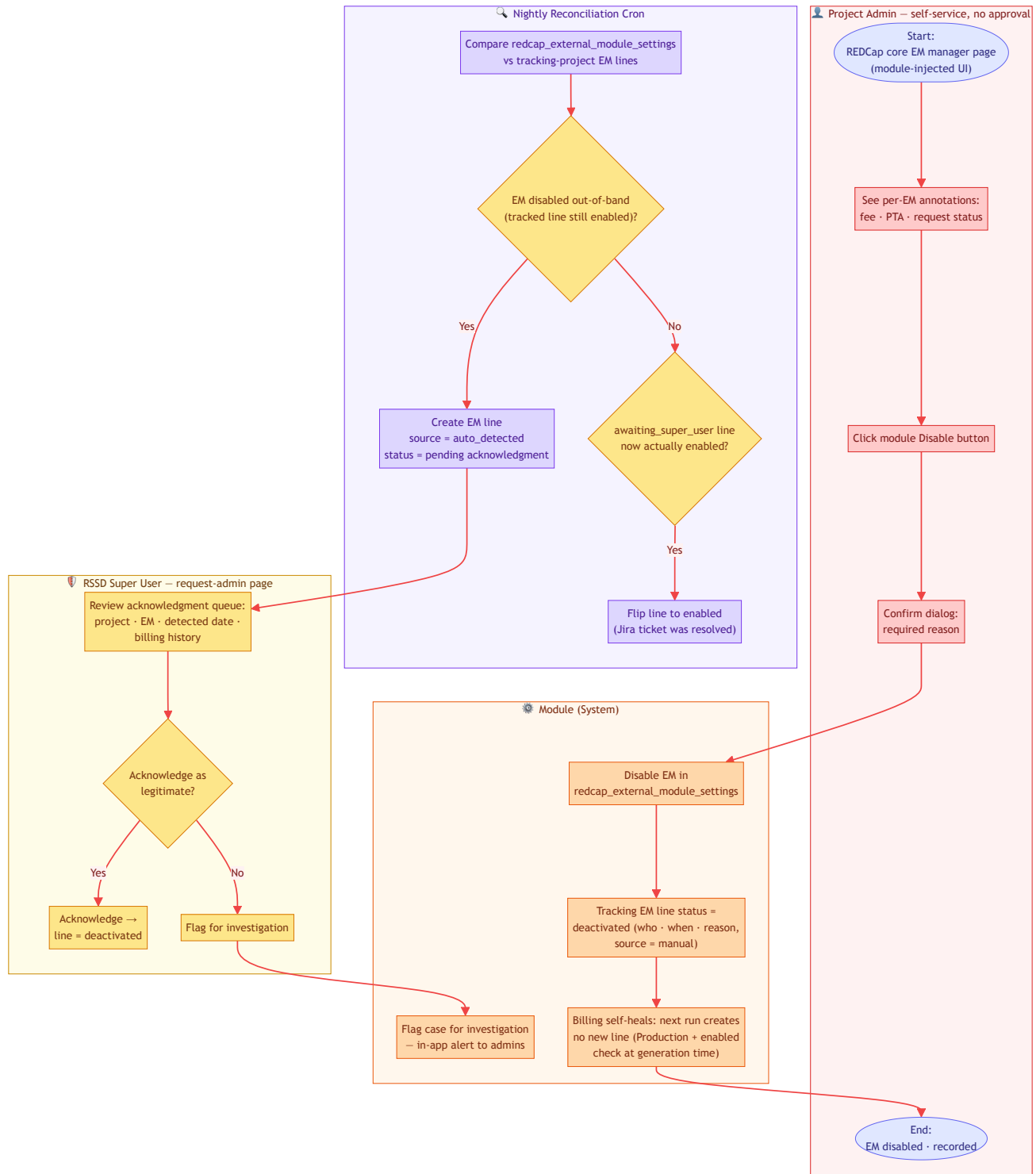
3) Diagram 1 — EM Enablement Request: Cart → Survey → PI Approval

Replaces the super-user approval queue. Requester (Design right) → cart → PTA checkout → tracking-project survey → Alert to PI → whole-request decision → hook: PTA sync, auto-enable or one RSSD Jira ticket.



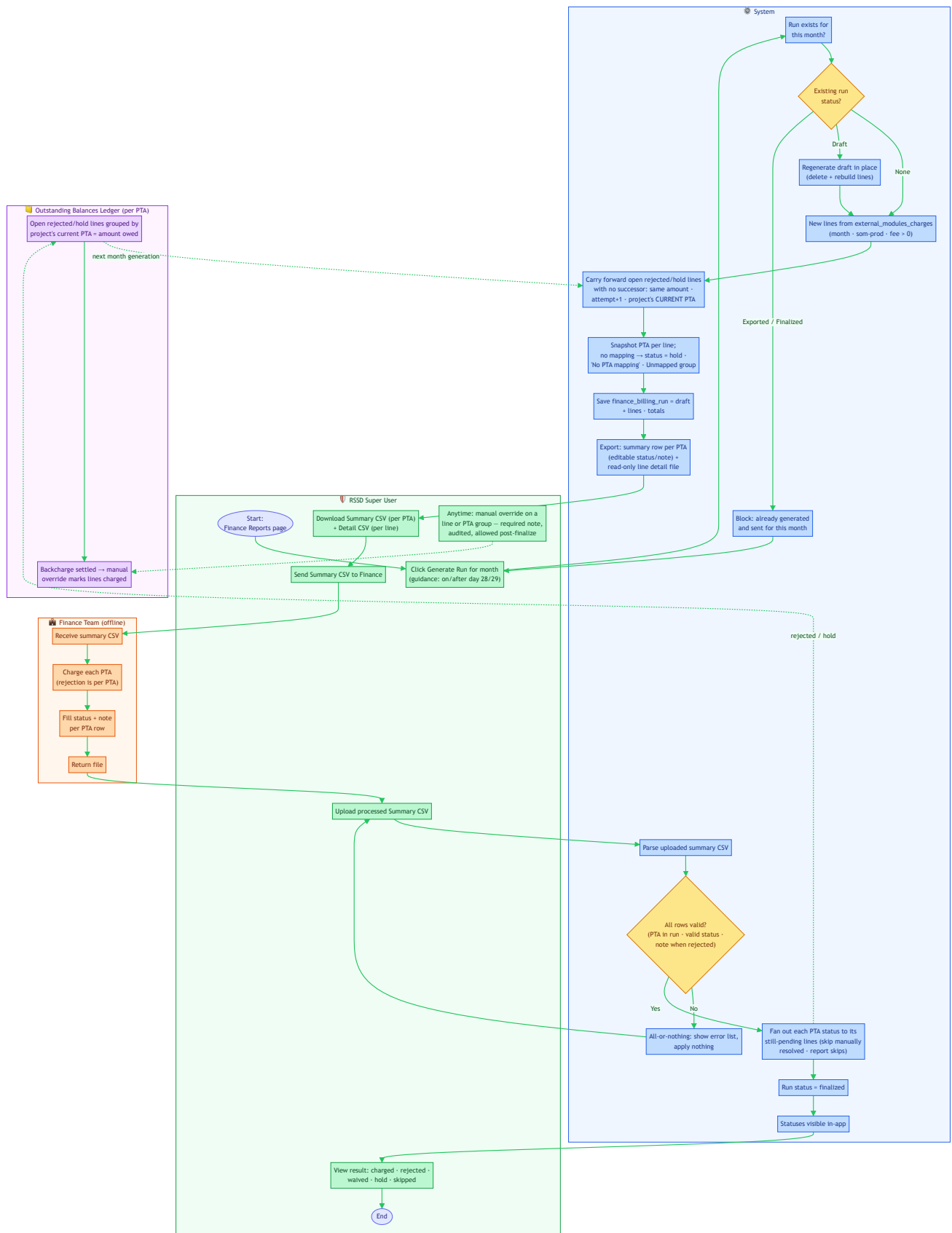
4) Diagram 2 — EM Deactivation & Nightly Reconciliation

Self-service Disable button injected into REDCap's core EM manager page (required reason, audited). Reconciliation files out-of-band disables for acknowledgment and closes Jira-resolved `awaiting_super_user` lines. Grandfather seeding establishes the go-live baseline.



5) Diagram 3 — Monthly Billing Run, Finance CSV Loop & Debt Ledger

One run per month (draft regenerates in place; exported/finalized blocks). Per-PTA summary CSV is the editable file; import fans statuses out to lines. Open rejected/hold lines carry forward on the project's current PTA and feed the PTA-keyed backcharge ledger.



6) Data Model

New entities (REDCapEntity)

- **finance_billing_runs** — run_month, run_year, status (draft/exported/finalized), generated_by, exported_at, imported_at, total_amount, total_lines, carried_amount.
- **finance_billing_lines** — run_id, instance, project_id, project_title, module_prefix, amount, pta_id / pta_number (snapshot; empty = unmapped), status, finance_note, status_source (generation/csv_import/manual), status_updated_by/at, carried_from_line_id, attempt_number, charge_entity_id.

Removed / superseded

- **monthly_pta_reports** — JSON snapshot can't track per-line status.
- **em_enable_requests** / **em_enable_request_items** — superseded by the request-tracking project's records.

Reused / extended

- **pta_accounts**, **project_pta_mapping** — kept current by approval-time PTA sync.
- **external_modules_charges** — monthly charge source (hub-synced); later extended with **source** + R2P2 traceability for the historical migration.
- EM-records project (PID 16000) — fee fields + new **auto_enable** / **requestable** flags.
- **Request-tracking project** — instruments: request survey (requester + person-lookup approver), approval survey (whole-request decision), repeating EM line (prefix, fee/auto-enable snapshots, **line_status**, Jira key, **source** = request/grandfathered/auto_detected).

7) Status Dictionaries

Request EM-line statuses

pending → approved → enabled | awaiting_super_user (→ enabled via reconciliation) | approval_error (retryable) · pending → rejected · enabled → deactivated

Billing line statuses

Status	Meaning	Next behavior
pending	Awaiting finance result	Await import / manual override
charged	Finance completed charge	Closed; no carry-forward

Status	Meaning	Next behavior
rejected	PTA charge failed	New line next month — same amount, attempt+1, project's <i>current</i> PTA
waived	Charge forgiven	Closed; no carry-forward
hold	Finance hold / no PTA mapping	Carries forward like rejected

Run statuses

draft → exported → finalized . One run per month; draft regenerates in place, exported/finalized blocks regeneration.

Finance CSV (summary — the editable file)

One row per PTA: `billing_run_id · billing_month · pta_number (re-import key) · pta_description · contact_name · project_count · line_count · new_amount · carried_amount · total_amount · status · finance_note` . Detail file (read-only): `line_id · pta_number · project_id · project_title · module_prefix · amount · attempt_number · carried_from_line_id · status` .

8) Files Touched

- `ExternalModuleManager.php` — entity types, billing methods (replace the PTA report section), save-record hook, every-page hook, reconciliation cron, Jira client.
- `config.json` — **no-auth fix**, new links (`em_catalog` , `request_admin`), settings (`em-request-project-id` , `pta-regex` , `jira-*`), `em_reconciliation` cron, hook permissions.
- `ajax/finance_report_action.php` · `ajax/finance_report_download.php` · new `ajax/em_request_action.php` .
- `pages/finance_reports.php` · new `pages/em_catalog.php` · new `pages/request_admin.php` (+ js/css counterparts).
- **Outside the repo:** request-tracking REDCap project (+ Alert + person-lookup EM), PID 16000 field additions, Jira service account.

Open items

- Confirm person-lookup EM = `stanford_person_lookup_v9.9.9` and its survey usage.

- Verify ExternalModules framework enable-API signature for the running REDCap version.
- RSSD Jira REST credentials + required issue fields.

9) Verification (local REDCap docker)

1. **Billing:** seed PTAs/mappings/charges → generate run (unmapped → hold/Unmapped) → download both CSVs → edit summary (one PTA charged, one rejected + note) → import → verify fan-out, finalize, re-import blocked, duplicate-generate blocked.
2. **Carry-forward:** next month's run recreates rejected/hold lines (attempt 2, current PTA after changing a mapping); charged/waived don't reappear.
3. **Ledger:** totals match open lines; manual mark-paid (note) removes from ledger with audit fields.
4. **Requests:** cart checkout → record + survey redirect → Alert email → PI approve → auto-enable verified in project / Jira line `awaiting_super_user` → PTA mapping updated; reject and `approval_error` retry paths.
5. **Deactivation:** core page annotations + Disable (reason recorded); native out-of-band disable → reconciliation files acknowledgment; manual enable of Jira line → flipped to enabled; grandfather seeding counts.
6. **Security:** logged-out hit of `ajax/finance_report_action.php?action=list_pta` returns 403.